

TrES-3b

R-1934

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_140427 · created 2026-08-06T13:55:47+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456774.7724 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.031
Transit depth (Rp/Rs)^2	2.47 +/- 0.98 [%]
Orbital Inclination (inc)	82.66 +/- 1.44
Airmass coefficient 1 (a1)	0.984 +/- 0.018
Airmass coefficient 2 (a2)	0.0129 +/- 0.0047
Scatter in the residuals of the lightcurve fit is	1.89 %
Best Comparison Star	#2 - [228, 389]
Optimal Aperture	3.45
Optimal Annulus	11.11
Transit Duration (day)	0.059 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.157 ± 0.031 vs 0.16309 (deviation 0.2σ)
Duration fit vs published	0.059 d vs 0.0567 d (deviation 0.18σ)
Mid-time O–C	-0.6 min
Depth SNR	5.1
Residual scatter	1.89 %

Light curve

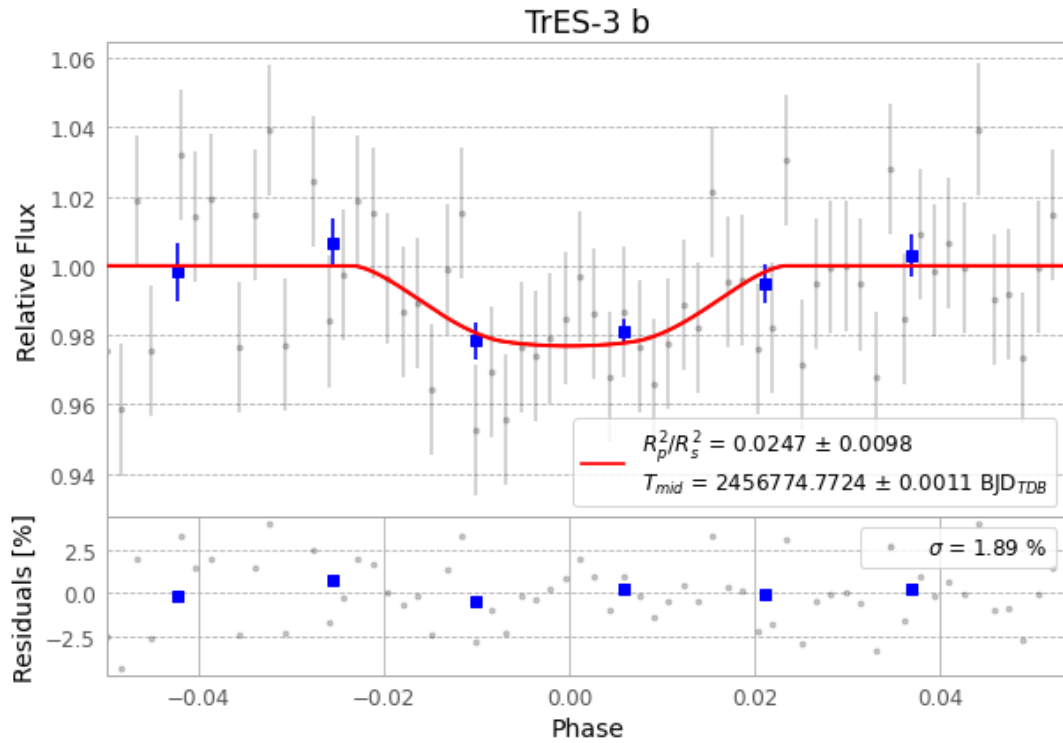


Figure 1 — FinalLightCurve_TrES-3 b_2014-04-26.png (SHA-256 4f44c90726904854...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [180, 179], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 63a3513b1aa4615e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit d8a7143

Data provenance

66 FITS frames (43 MB), TRES-3140427045412.FITS → TRES-3140427080912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-140427.log (b2177c011206...), FinalLightCurve_TrES-3 b_2014-04-26.png (4f44c9072690...), FinalLightCurve_TrES-3 b_2014-04-26.csv (4cff0d849ba9...)

Compute resources

Wall time 102.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 13:55Z.